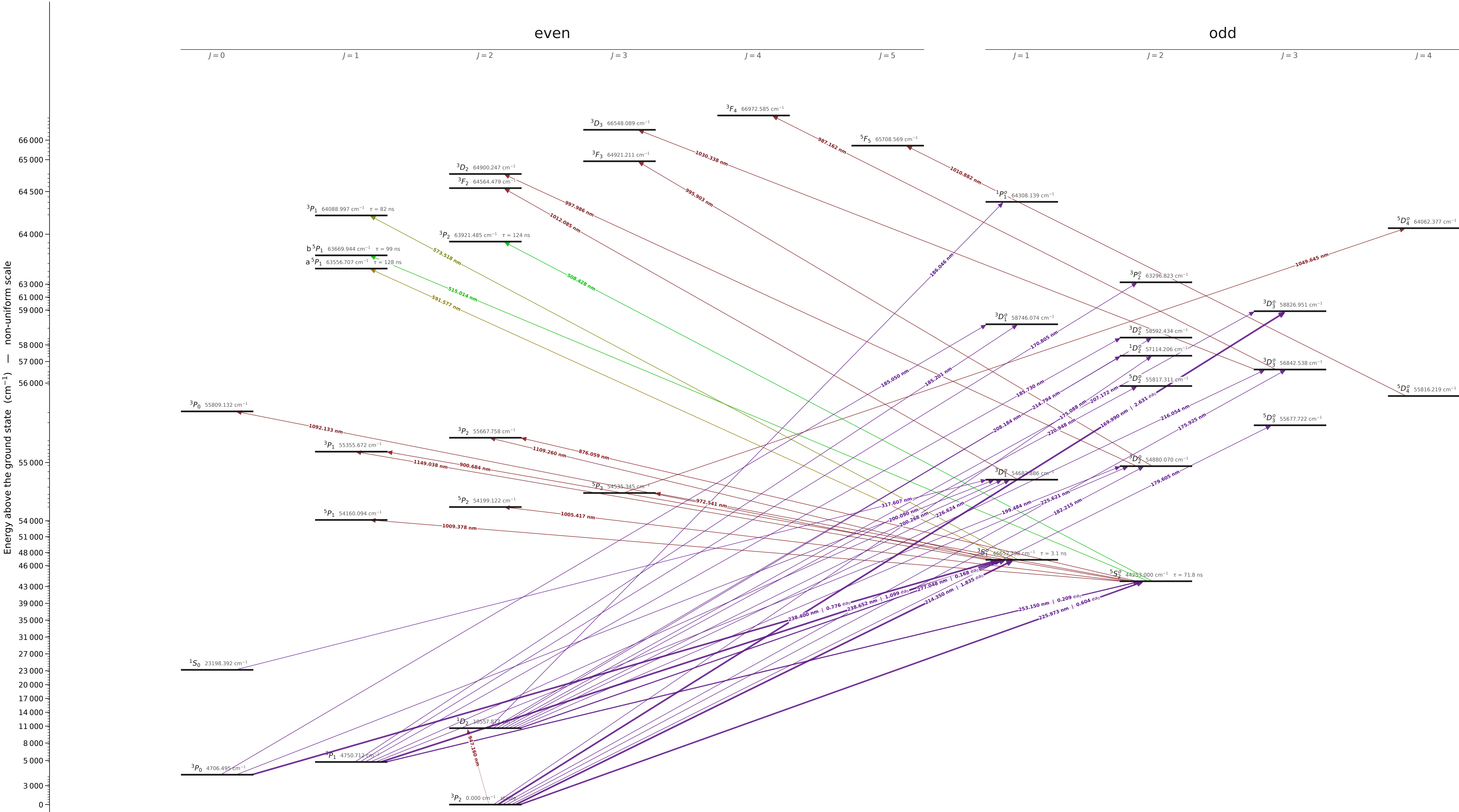
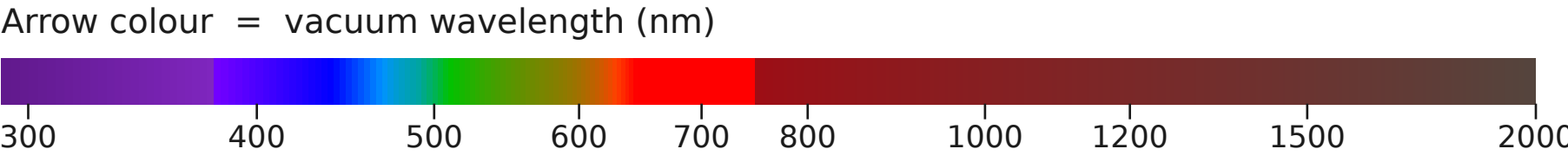


Tellurium

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e r J' \rangle $ ($e a_0$)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$^3P_2 - ^3S_2^o$	214.3497	214.2822	1398.6139	1.835	2.310e+08	51.34	71.6	first step of the ISOLDE RILIS resonance-ionization schem...
$^3S_2^o - ^3P_1$	573.5175	573.3585	522.7259	-	-	1.941	-	second step of the ISOLDE RIUIS resonance-ionization sche...
$^3S_1^o - a\ ^5P_1$	591.5770	591.4132	506.7682	-	-	1.243	-	second resonant step

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

¹²⁵Te hyperfine structure ($I = 1/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3P_2	-998(14)	-	F=3/2: +1497.00 F=5/2: -998.00
$^5S_2^o$	1010.4(13)	-	F=3/2: -1515.60 F=5/2: +1010.40
$^3S_1^o$	943(20)	-	F=1/2: -943.00 F=3/2: +471.50

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^3P_2 - ^3S_1^o$	214.350	120-130	-1029(11)
$^3P_2 - ^3S_1^o$	214.350	122-130	-829(13)
$^3P_2 - ^3S_1^o$	214.350	124-130	-625(13)
$^3P_2 - ^3S_1^o$	214.350	126-130	-426(11)
$^3P_2 - ^3S_1^o$	214.350	128-130	-211(11)
$^3P_2 - ^3S_1^o$	214.350	132-130	214(15)
$^3P_2 - ^3S_1^o$	214.350	134-130	402(15)
$^3P_2 - ^3S_1^o$	214.350	136-130	1242(21)
$^3P_2 - ^3S_1^o$	214.350	123-130	-815(15)
$^3P_2 - ^3S_1^o$	214.350	125-130	-594(19)

Shift = ν (first isotope) - ν (second isotope).